DFMEA - VBF-T5R2-DFMEA-01

Case t5_r2 finalist f2_h70_C. Qualitative S(1-5)/O(1-5) scoring (engineering judgment), RPN = S x O, current controls cite actual simulation signals where they exist. This is a design-stage DFMEA; physical failure statistics are not available in-simulation.

#	Function / failure mode	Effect	S	Cause	O	Current detection / control (simulation signal)	RPN	Recommended action
---	---	---	---	---	---	---	---	---
1	Deliver 4C charge without Li plating	dendrite risk, active Li loss	5	anode-side transport insufficient at 26 A (low t+ electrolyte, low porosity, coarse particles)	2	4C_charge_45C anode potential min +0.0506 V (51 mV margin) - engineered by transport-first order of levers	10	monitor in pack cells; maintain t+ >= 0.5 electrolyte spec
2	Keep T <= 60 C at 4C	electrolyte degradation, venting	4	cooling-path degradation (fouling, dry-out) or channel underflow	2	lumped T_max 330.984 K vs 333.15 in 4C_charge_45C; margin 2.2 K	8	h >= 70 W/m2/K channel verification test; sensor-in-pack DVP
3	Meet ED >= 500.94 Wh/kg class	falls below flagship target	3	mass creep in manufacture (collector/coating over-tolerance)	2	calc-energy mass ledger (38.083 g contract, 50.511 g BOM) per layer	6	SPC on coating & foil thickness
4	Maintain capacity over life	reduced range	3	SEI growth (ec-reaction-limited kinetics)	3	aging_1C_100cyc end-SEI 520.3 nm (informational, artifact-flagged)	9	verified aging campaign + post-mortem; formation per recommended program
5	Mechanical integrity of 98/122 um electrodes	delamination, capacity loss	4	insufficient adhesion at high compaction (1.86/0.91 g/cm3)	2	none in-sim (out of simulator scope)	8	adhesion & bending tests; binder optimization (98/1/1)
6	Electrolyte dry-out / leak	ionic starvation, safety	4	seal failure at high T cycles	2	none in-sim	8	leak/thermal-humidity qualification tests
7	Fast-charge voltage overshoot	4.2 V exceedance, electrolyte oxidation	3	CC-only profile without taper in real BMS	2	protocol cut-off logic (simulation enforces 4.2 V)	6	tapered charge profile in BMS (see design_spec 7)

Highest RPN = #1 (S5 x O2 = 10) and #4 (S3 x O3 = 9). All scoring explicitly qualitative.